Personal Learning Journey

IT3091 Machine Learning - Group 2026-AI-05

DataCo Smart Supply Chain: late-delivery risk

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Area of responsibility: Exploratory data analysis and preprocessing

My role

I worked on the exploratory data analysis and then owned the preprocessing pipeline. That meant taking the raw DataCo file of 180,519 rows and 53 columns and turning it into something a model could actually learn from, and being able to justify every column I removed.

What I did

I started by reading every column against the supplied data dictionary and building an audit table of types, distinct values and missing rates. From there I produced 31 figures covering class balance, delay rates by shipping mode, market, region and category, and the distribution of every numeric column.

The preprocessing itself involved removing four leakage columns, filtering out 7,754 cancelled orders, designing the train/validation/test split, and building a scikit-learn pipeline that imputes, scales and one-hot encodes 26 features into 132 columns. I also tested feature selection and PCA.

What I learned

The biggest thing I learned is that leakage is not something you assume, it is something you prove. I suspected the target was just a calculation on two other columns, so I tested it: the rule real shipping days greater than scheduled days reproduces the label on 97.55% of rows. Once I had that number, dropping those columns was an easy decision to defend.

I also learned why the splitting strategy matters more than I expected. Our rows are order line items, not orders, so one order can span five rows that all share the same label. A random split would have put the answer on both sides. Switching to a chronological split fixed that and also matched how the model would really be used, which is on future orders.

On PCA, I learned to test rather than assume. I expected it to help because the feature count looked high, but at the same number of inputs feature selection beat it by 0.020 ROC-AUC and stayed interpretable. Most of our columns are one-hot binaries, which PCA handles poorly, and it optimises variance rather than relevance.

What I found difficult

Deciding when a column was genuinely redundant was harder than it looked. I found that Order Item Total equals Product Price times Quantity minus Discount, exactly, on every row, but the correlation between those columns never went above 0.91 because the relationship is multiplicative. If I had only looked at a correlation matrix I would have missed it completely.

I also found it uncomfortable to accept that our best model would only reach about 0.78. My instinct was that a better model would fix it, but the EDA showed one feature carries almost all the signal, so the ceiling is a property of the data.

A mistake I caught

I originally wrote that the unseen product categories in the test set were routed into an infrequent bucket by the encoder. When I checked the actual encoder output I found that was wrong: the rare-level threshold

never fires on our data because every training level already has at least 20 rows, so unseen levels are encoded as all zeros instead. It did not change the result, but it would have been a wrong statement in the report.

What I would do differently

I would do the leakage audit before any EDA rather than alongside it. I drew several charts using columns that I later had to drop, which wasted time. I would also check for exact arithmetic relationships between columns earlier, since a correlation matrix on its own does not find them.

On using AI

I used Claude to write and debug code and to suggest checks I had not thought of. It was genuinely useful, but it was also confidently wrong several times, and the only way I caught those was by running the code and reading the output rather than the explanation. That is the habit I am taking from this assignment.